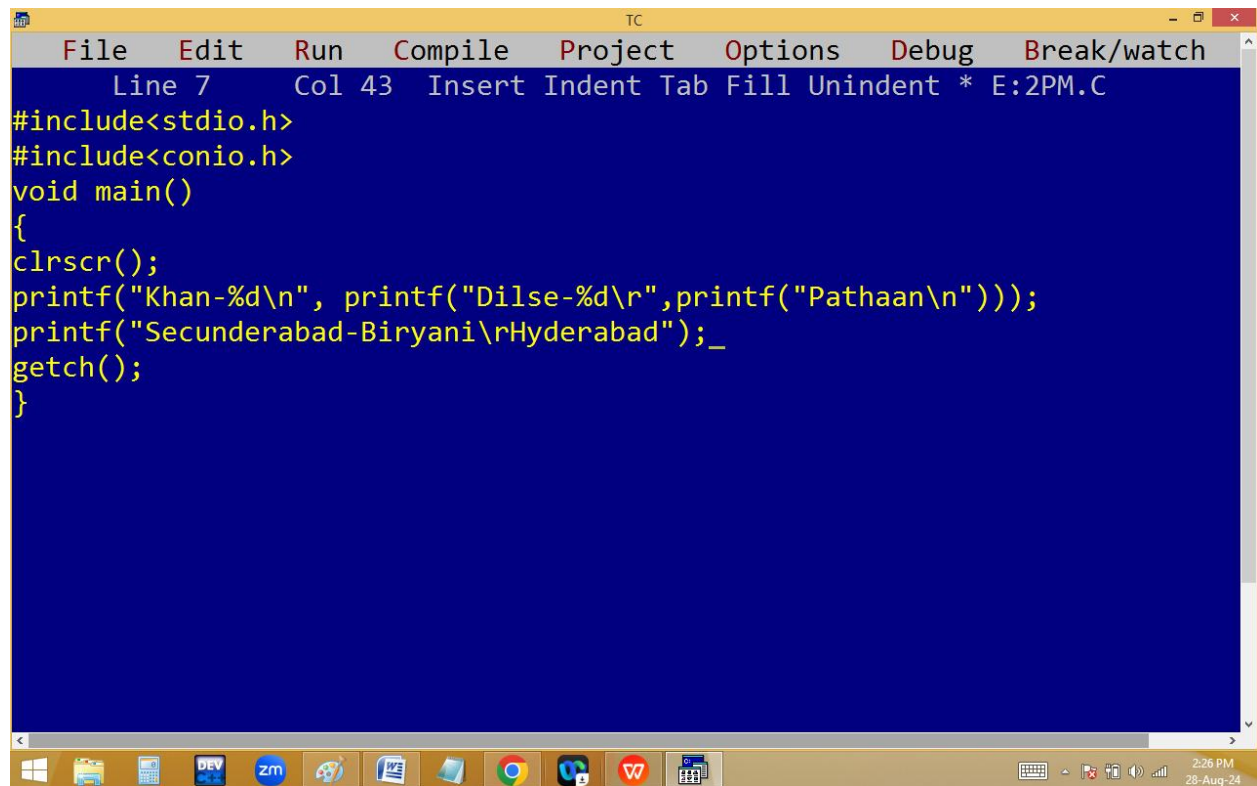




```
File Edit Run Compile Project Options Debug Break/watch
Line 7 Col 43 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
printf("Khan-%d\n", printf("Dilse-%d\r",printf("Pathaan\n")));
printf("Secunderabad-Biryani\rHyderabad");_
getch();
}
```

```
TC
Pathaan
Khan-88
Hyderabad-Biryani
```

p("Pathaan\n");

8

8

p("Dilse-%d\r");

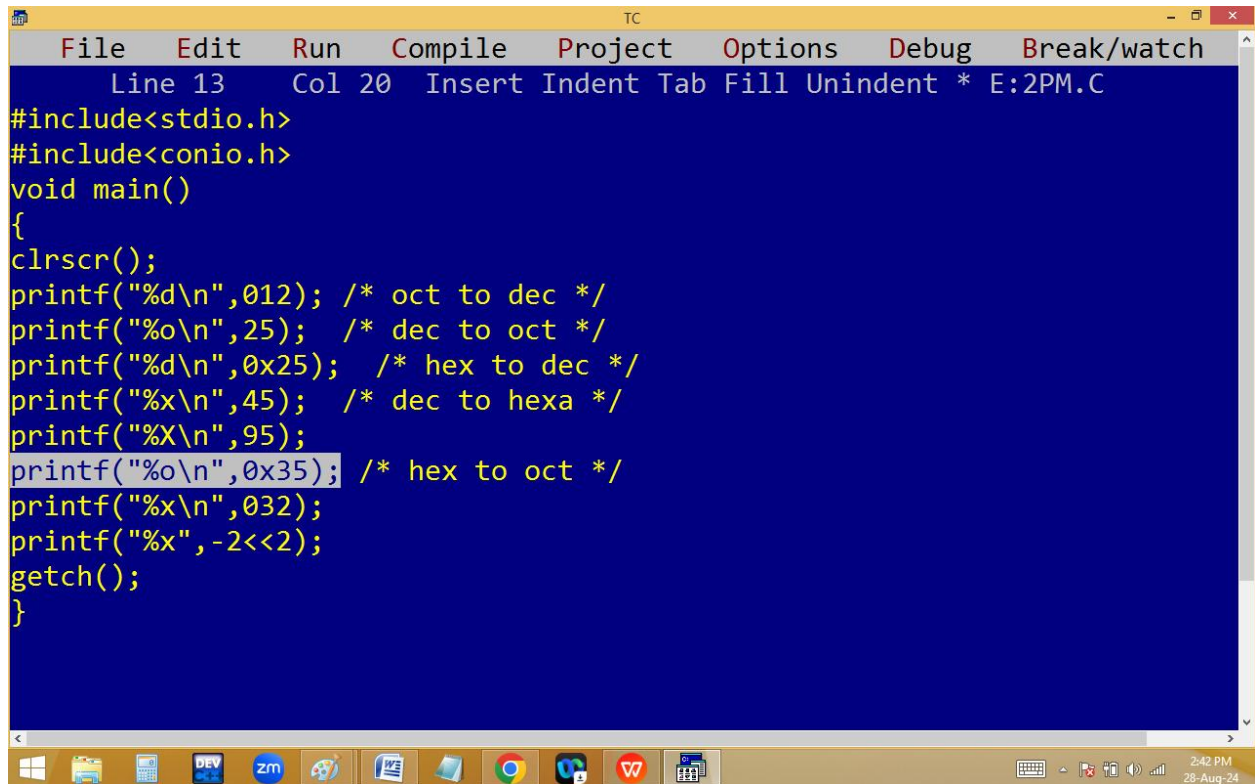
8

8

p("Khan-%d");

Pathaan
~~Dilse-8~~
Khan-8

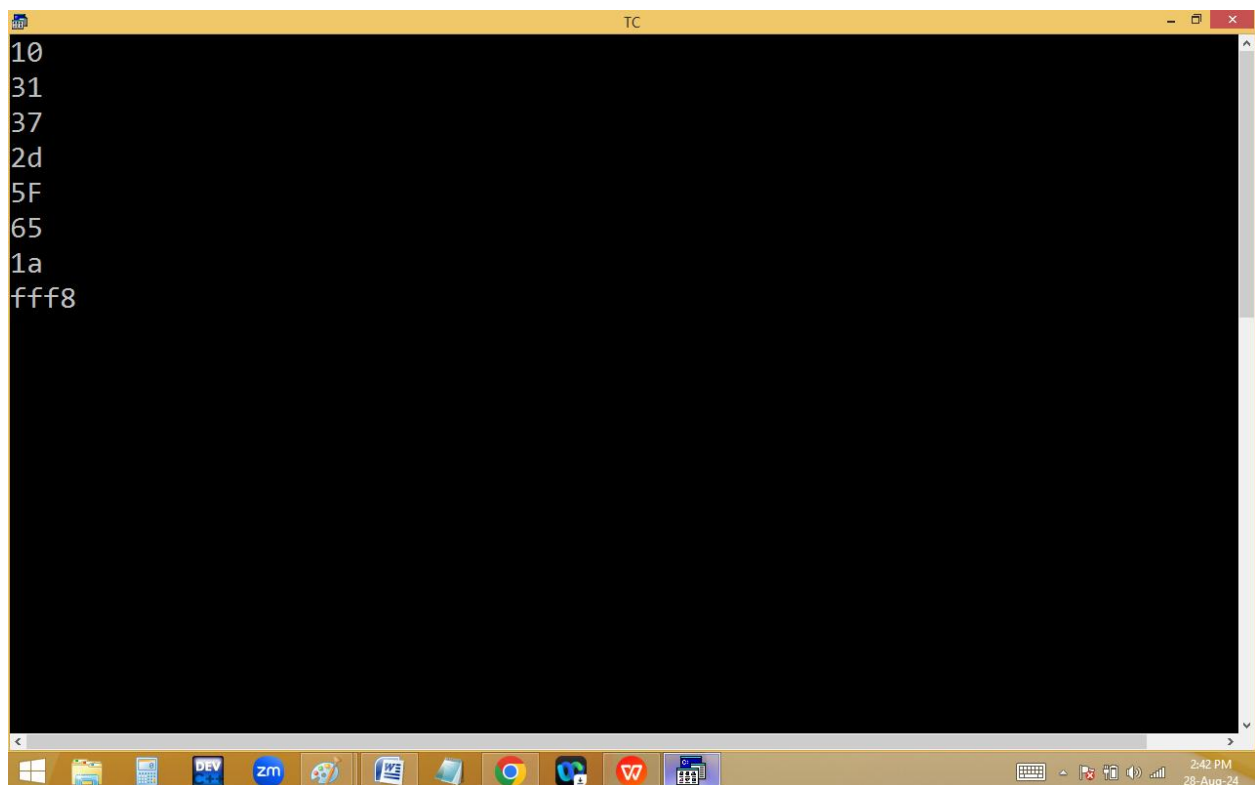
~~Secunderabad-Biryani\r~~
Hyderabad



The screenshot shows the Turbo C++ (TC) IDE with a menu bar (File, Edit, Run, Compile, Project, Options, Debug, Break/watch) and a status bar (Line 13, Col 20, Insert, Indent, Tab, Fill, Unindent, * E:2PM.C). The code in the editor is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
printf("%d\n",012); /* oct to dec */
printf("%o\n",25); /* dec to oct */
printf("%d\n",0x25); /* hex to dec */
printf("%x\n",45); /* dec to hexa */
printf("%X\n",95);
printf("%o\n",0x35); /* hex to oct */
printf("%x\n",032);
printf("%x",-2<<2);
getch();
}
```

The taskbar at the bottom shows various application icons and the system clock indicating 2:42 PM on 28-Aug-24.



The screenshot shows the Turbo C++ (TC) IDE with the same menu bar and status bar. The output window displays the results of the program's execution:

```
10
31
37
2d
5F
65
1a
fff8
```

The taskbar at the bottom is identical to the first screenshot, showing the same application icons and system clock (2:42 PM on 28-Aug-24).

oct to dec

$$\begin{array}{r} 012 \\ | \quad \diagdown \\ 8^1 \times 1 + 8^0 \times 2 \\ \hline 8 + 2 = 10 \end{array}$$

dec to oct

$$\begin{array}{r} 8 \overline{) 25} \\ 3 - 1 \checkmark \end{array}$$

hex to dec

$$\begin{array}{r} 0 \times 25 \\ | \quad \diagdown \\ 16^1 \times 2 + 16^0 \times 5 \\ \hline 32 + 5 = 37 \end{array}$$

dec to hex

$$\begin{array}{r} 16 \overline{) 45} \\ 2 - 13 \end{array}$$

0-9 10-a, 11-b, 12-c, 13-d, 14-e, 15-f

dec to hexa

$$\begin{array}{r} 16 \overline{) 95} \\ 5 - 15 \end{array}$$

hexa to dec

$$\begin{array}{r} 0 \times 35 \\ | \quad \diagdown \\ 16^1 \times 3 + 16^0 \times 5 \\ \hline 48 + 5 = 53 \end{array}$$

dec to oct

$$\begin{array}{r} 8 \overline{) 53} \\ 6 - 5 \checkmark \end{array}$$

oct to decimal

$$\begin{array}{r} 032 \\ | \quad \diagdown \\ 8^1 \times 3 + 8^0 \times 2 \\ \hline 24 + 2 = 26 \end{array}$$

dec to hexa

$$\begin{array}{r} 16 \overline{) 26} \\ 1 - 16 \\ 2 \checkmark \end{array}$$

$$-2 \ll 2 = -8$$

$$\begin{array}{r} 2 \overline{) 8} \\ 2 \overline{) 4} - 0 \\ 2 \overline{) 2} - 0 \\ 1 - 0 \end{array}$$

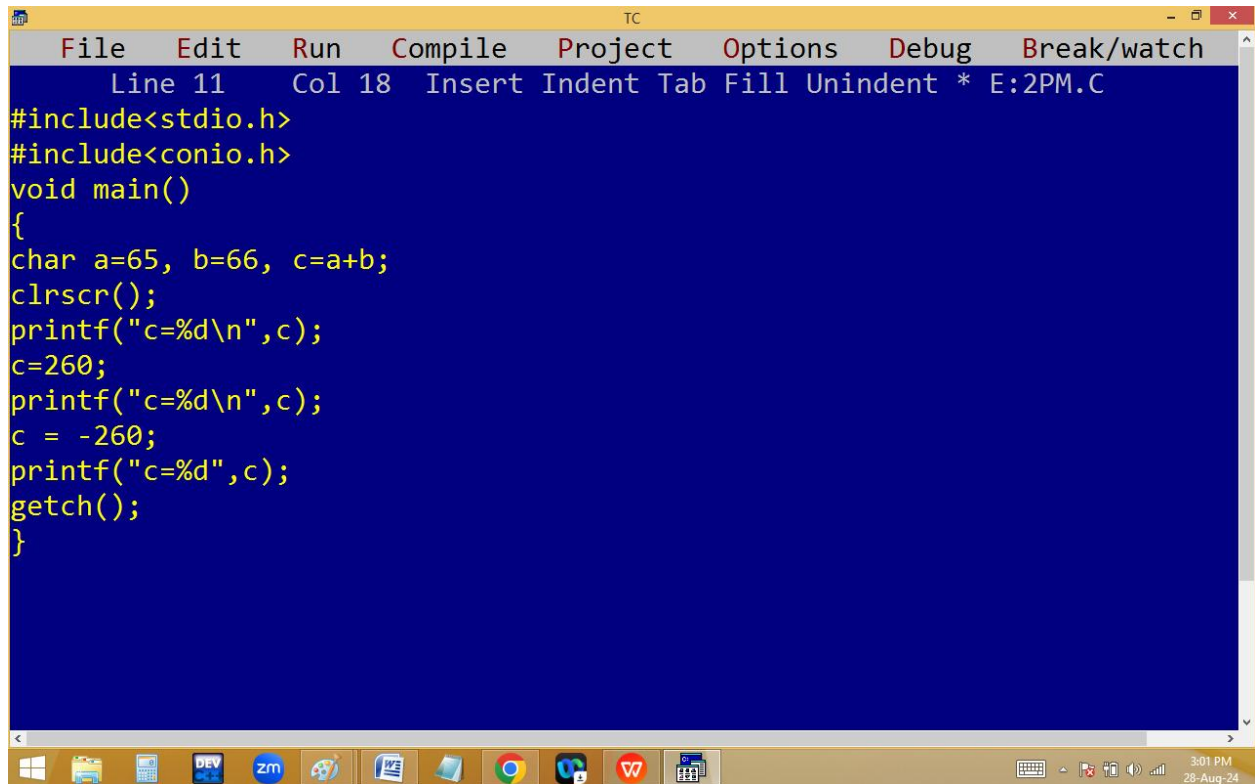
$$8 = 0000 \ 0000 \ 0000 \ 1000$$

$$1' \sim 1111 \ 1111 \ 1111 \ 0111$$

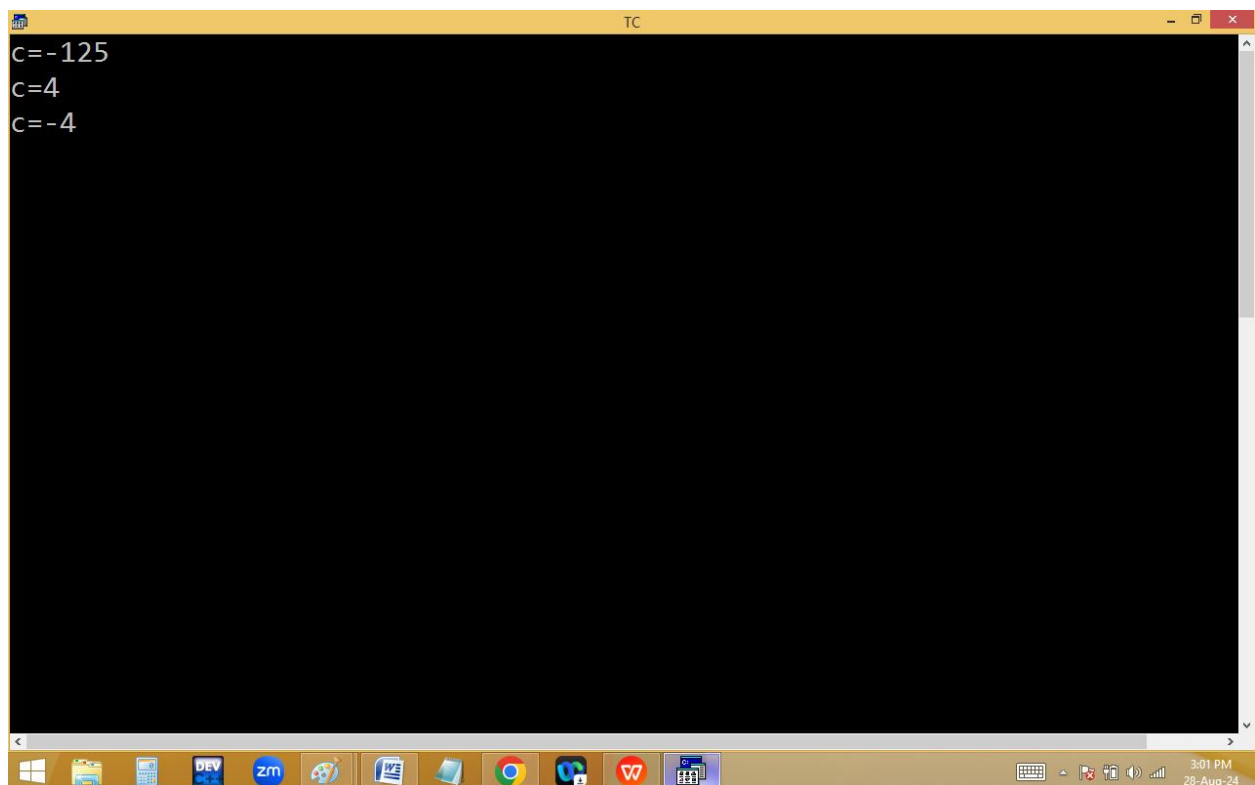
$$2' \sim 0000 \ 0000 \ 0000 \ 0001$$

$$1111 \ 1111 \ 1111 \ 1000$$

$$\begin{array}{r} \hline \text{f} \quad \text{f} \quad 15 = \text{f} \quad 8 \end{array}$$



```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 11 Col 18 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
char a=65, b=66, c=a+b;
clrscr();
printf("c=%d\n",c);
c=260;
printf("c=%d\n",c);
c = -260;
printf("c=%d",c);
getch();
}
```

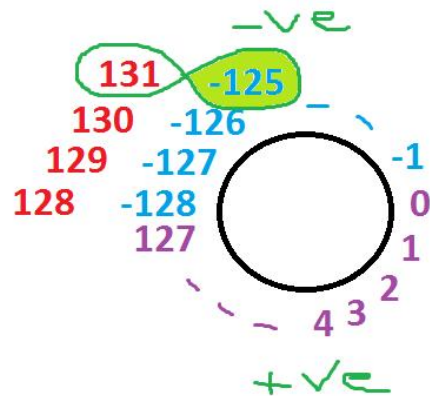


```
TC
c=-125
c=4
c=-4
```

c using ascii character set - 256 char's
signed char ==> -128 to + 127

$$\begin{array}{r} a=65 \\ b=66 \\ \hline 131 \end{array} \quad \begin{array}{r} 256 \\ -131 \\ \hline -125 \end{array}$$

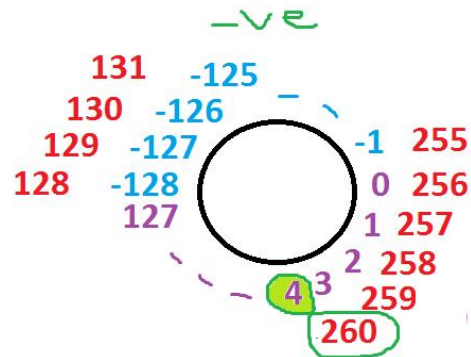
$$\begin{array}{r} 131 \\ -128 + 3 = -125 \\ \hline 3 \end{array}$$



c using ascii character set - 256 char's
signed char ==> -128 to + 127

$$c = 260;$$

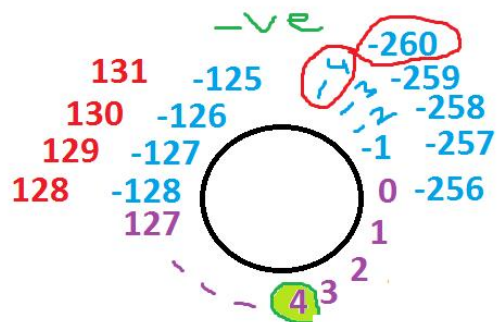
$$260-256=4$$

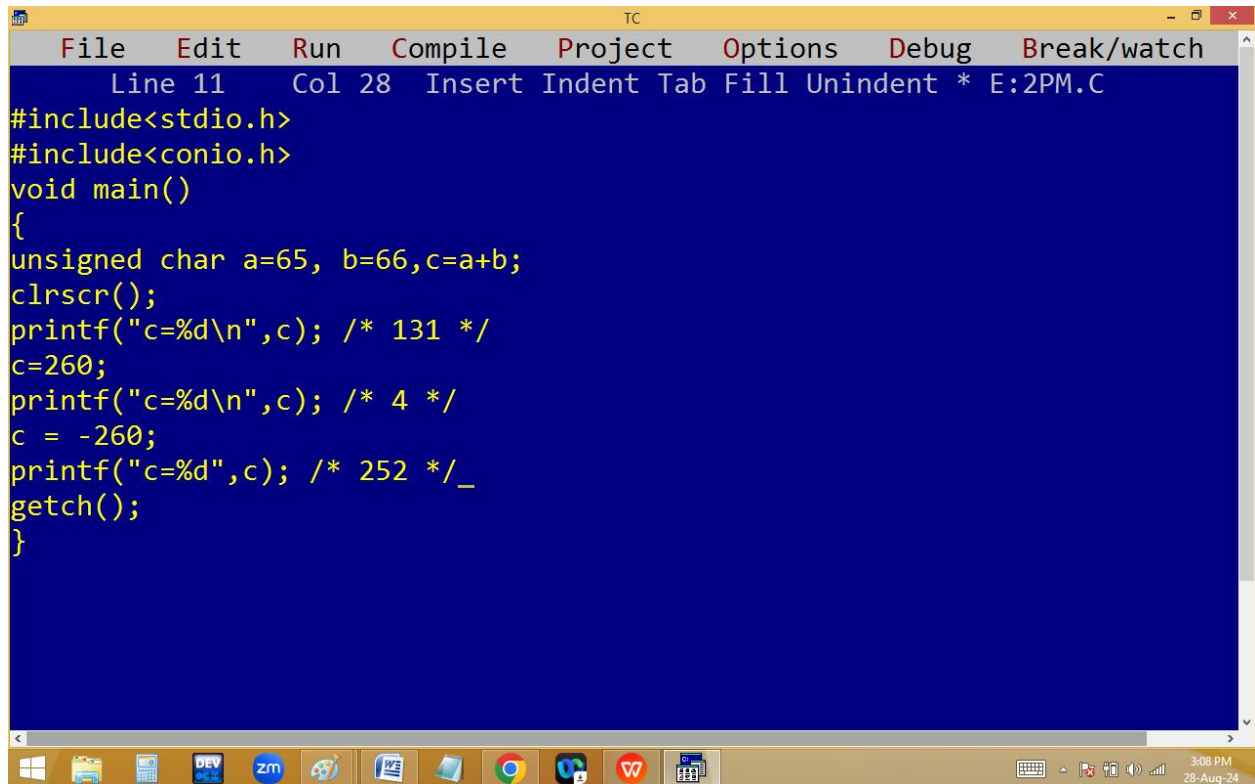


c using ascii character set - 256 char's
signed char ==> -128 to + 127

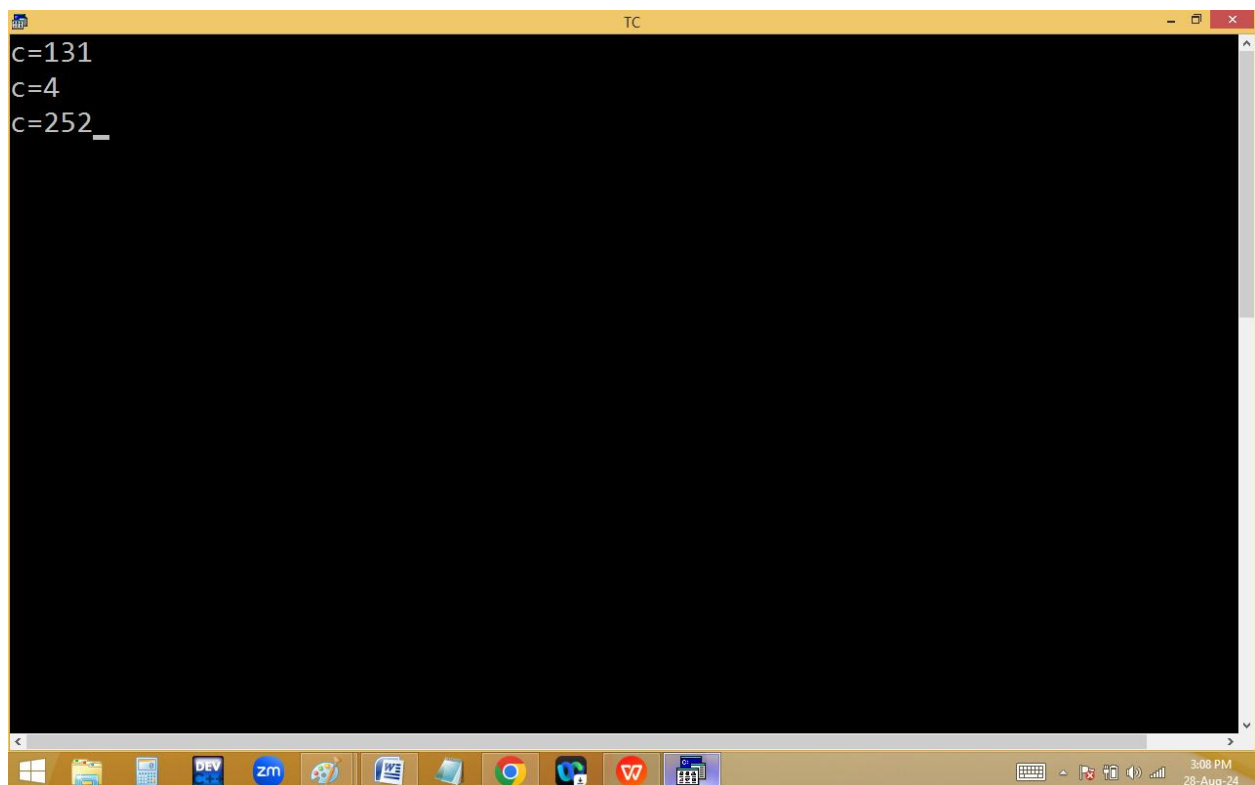
$$c = 260;$$

$$260-256=4$$





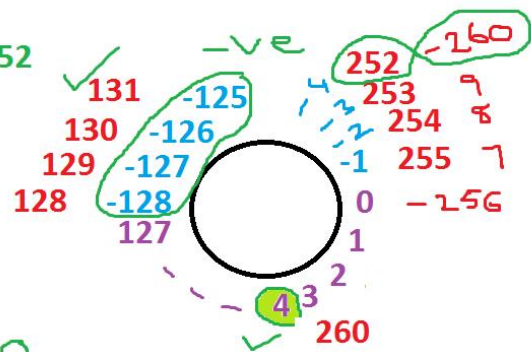
```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 11 Col 28 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
unsigned char a=65, b=66,c=a+b;
clrscr();
printf("c=%d\n",c); /* 131 */
c=260;
printf("c=%d\n",c); /* 4 */
c = -260;
printf("c=%d",c); /* 252 */_
getch();
}
```



```
TC
c=131
c=4
c=252_
```

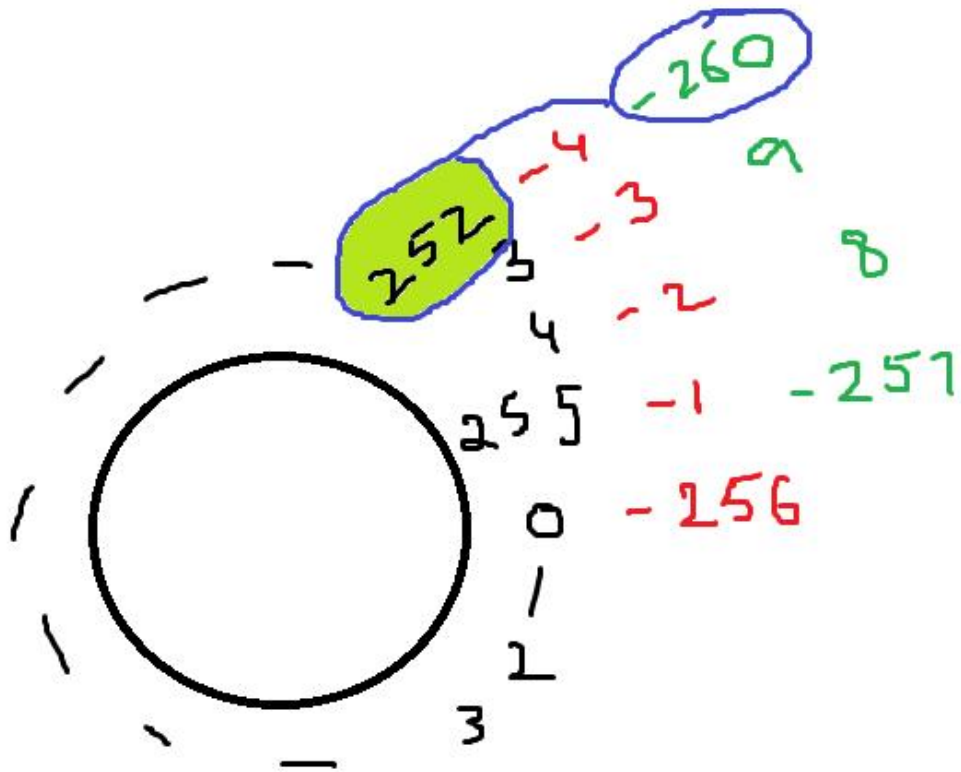
unsigned char ==> 0 to 255

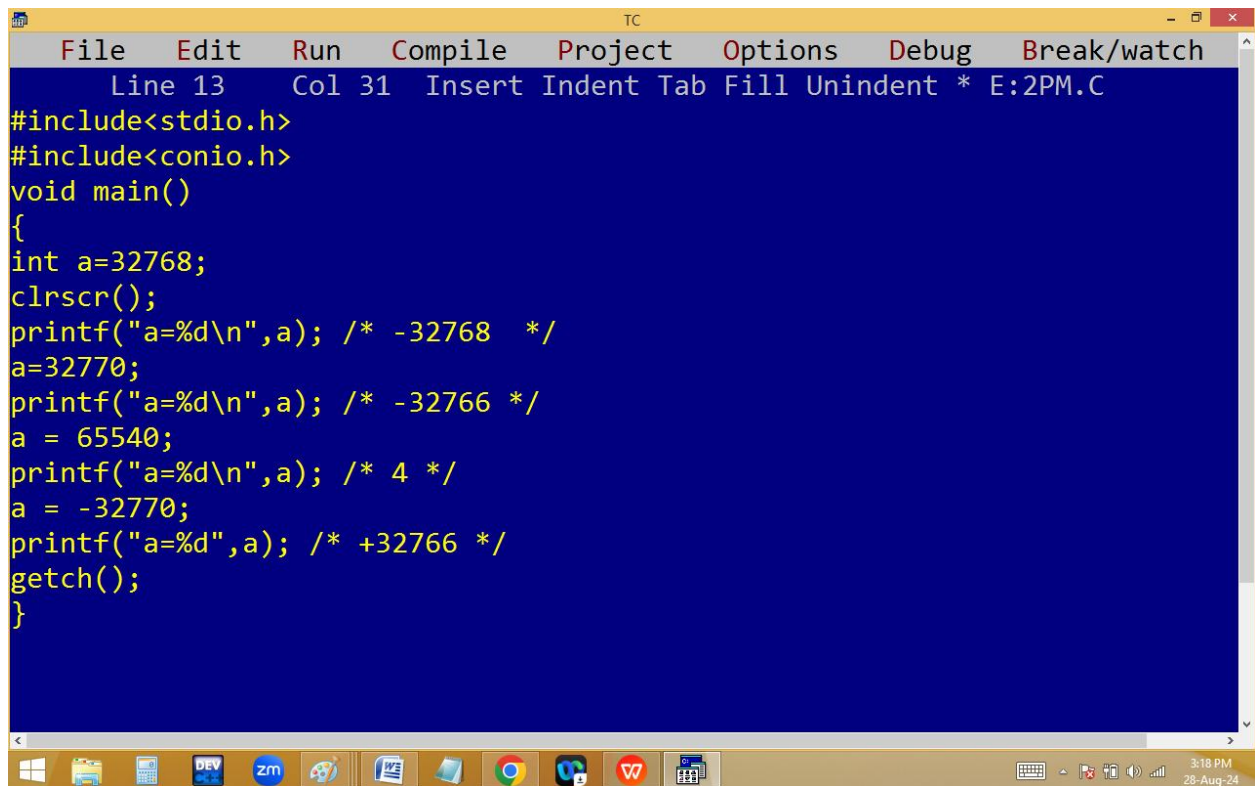
$$256 - 4 = 252$$



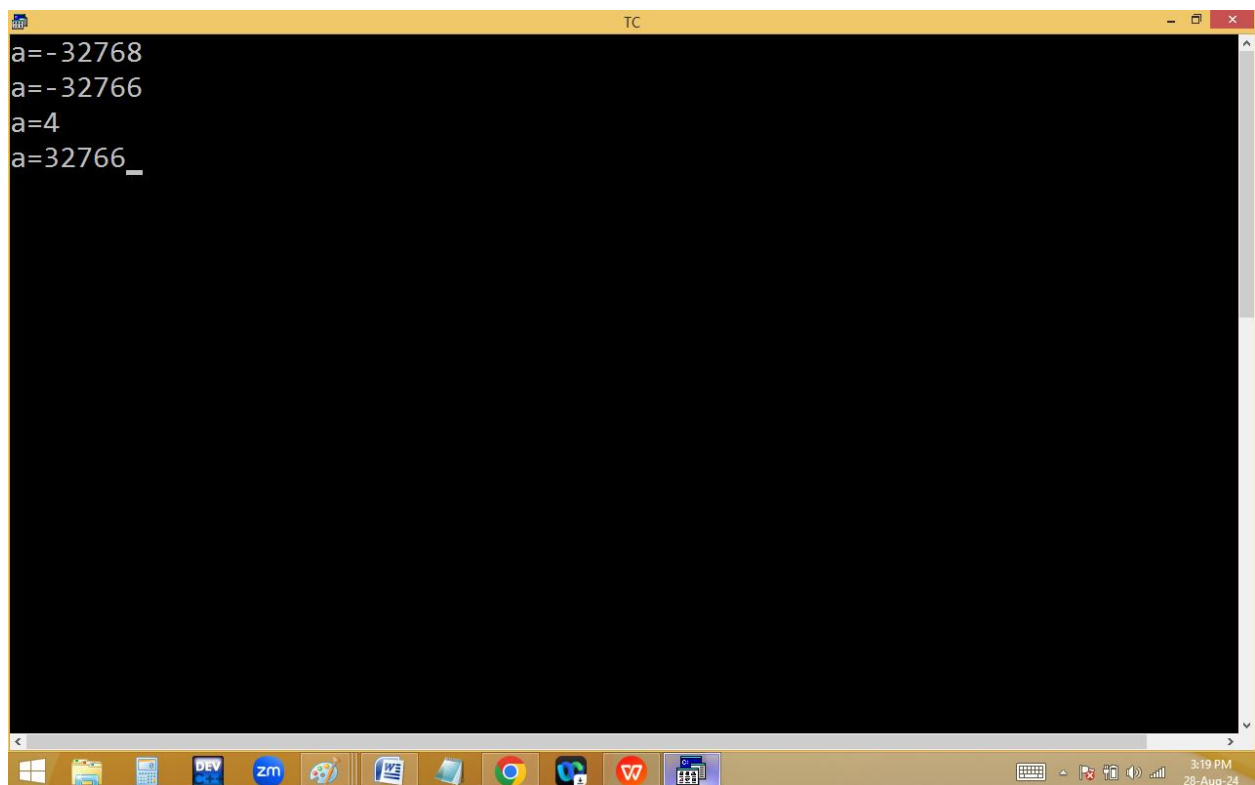
$$\begin{array}{r} 260 \\ -256 \\ \hline 4 \end{array}$$

$$\begin{array}{r} -260 \\ 256 \\ \hline 256 - 4 = 252 \end{array}$$





```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 13 Col 31 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int a=32768;
clrscr();
printf("a=%d\n",a); /* -32768 */
a=32770;
printf("a=%d\n",a); /* -32766 */
a = 65540;
printf("a=%d\n",a); /* 4 */
a = -32770;
printf("a=%d",a); /* +32766 */
getch();
}
```



```
TC
a=-32768
a=-32766
a=4
a=32766_

```

int - 16 bit compiler - $2^{16} \Rightarrow 65536/2=32768$

signed int [-32768 to +32767]

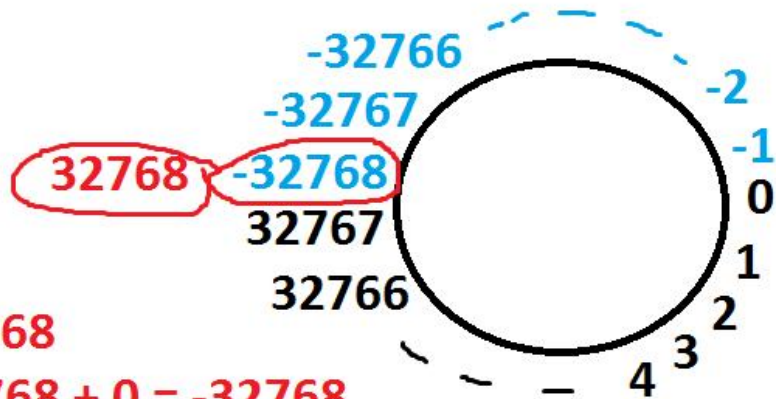
unsigned int [0 to 65535]

signed int cycle

$a = 32768$

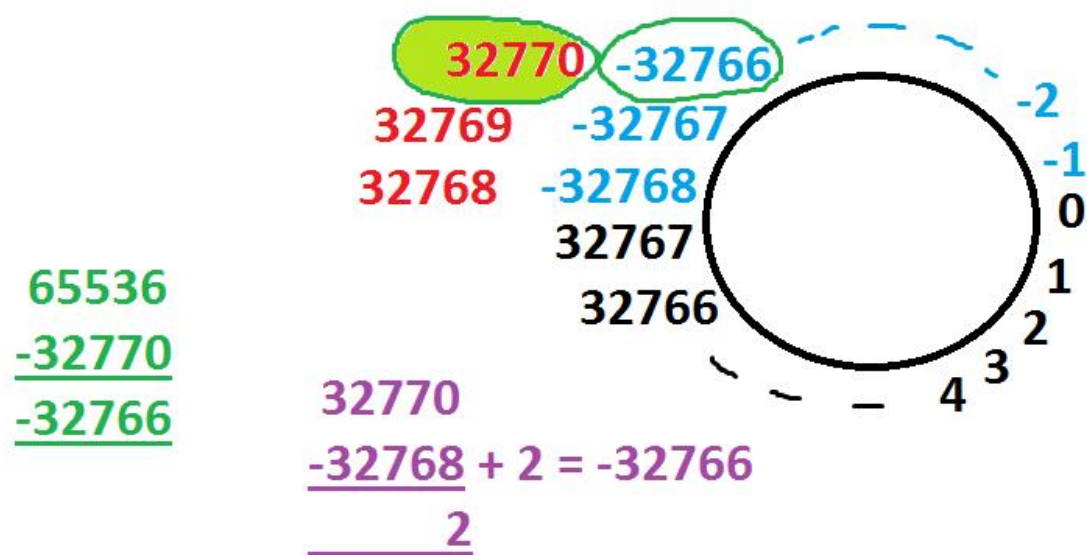
65536
32768
-32768

32768
-32768 + 0 = -32768
0

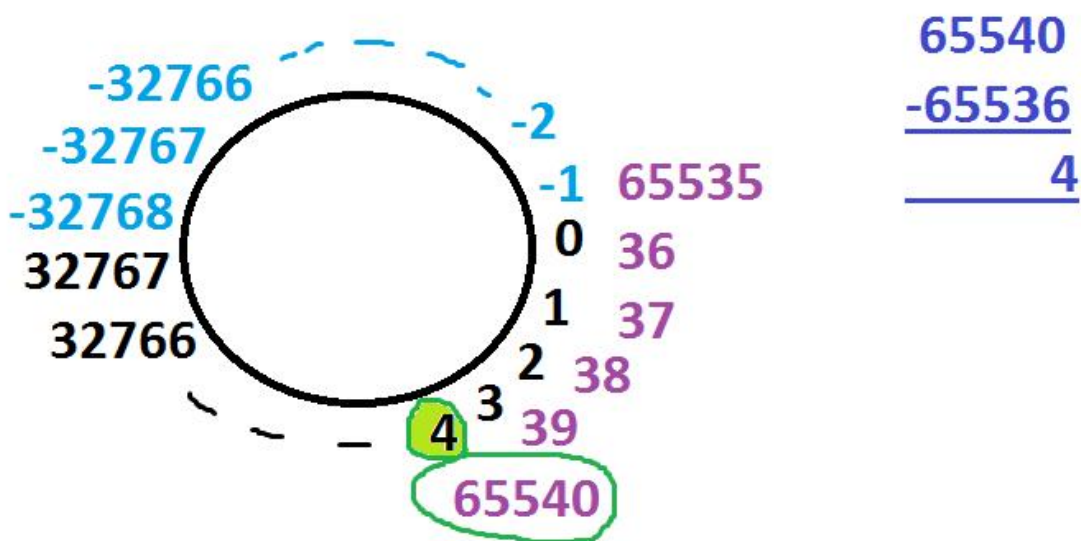


signed int cycle

a = 32768



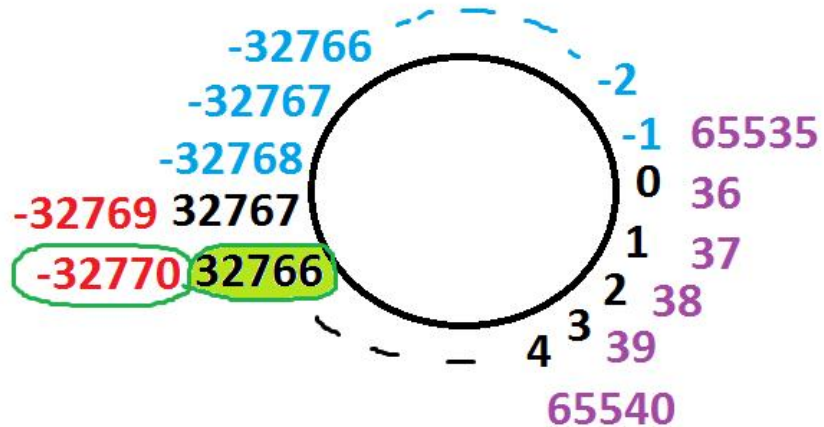
signed int cycle



signed int cycle

a = -32770

65536
-32770
32766



Variables:

Variable is a container is used to store the values in our programs.

Variable is a named memory location where we can store and manipulate [modify] the values in our programs.

In c compiler the variables should be declared in first line of any function. In c++ we can declare anywhere.

All the variables are temporary only because of they are stored in **primary memory** i.e. **RAM**. Once the program execution completed, automatically the variables are deleted from memory.

Variables are **case sensitive**. i.e. lower and upper are different.

```
Int a=100;
```

```
Int A=444;
```

```
Printf("a=%d, A=%d",a,A);
```

Output: a=100, A=444

Every variable having 2 stages.

1. Declaration / declared

Eg: **int a;**

2. Initialization / defined

a=100;

syntax:

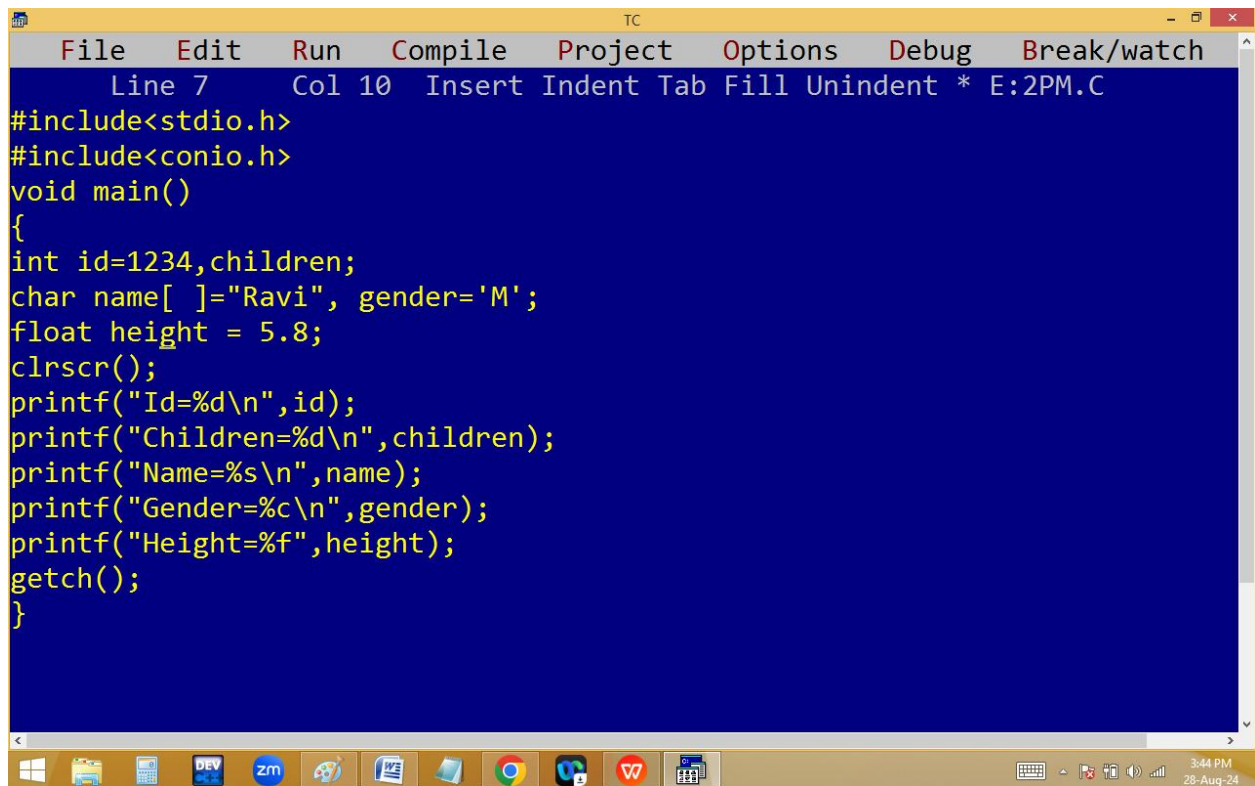
```
datatype variable =[value], variable[=value],.....;
```

Eg:

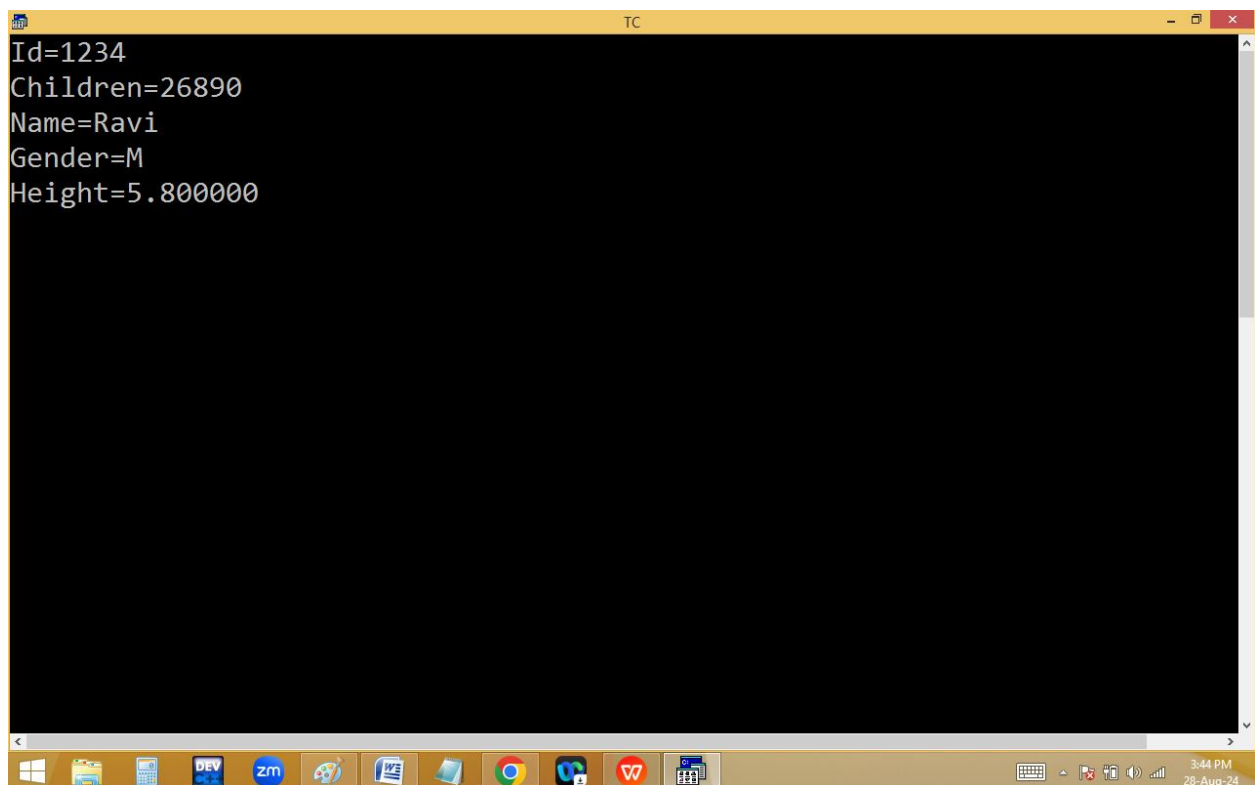
```
int id=1234, children=2;
```

```
char name[]="Ravi", gender='M';
```

```
float height = 5.8;
```



```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 7 Col 10 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int id=1234,children;
char name[ ]="Ravi", gender='M';
float height = 5.8;
clrscr();
printf("Id=%d\n",id);
printf("Children=%d\n",children);
printf("Name=%s\n",name);
printf("Gender=%c\n",gender);
printf("Height=%f",height);
getch();
}
```



```
TC
Id=1234
Children=26890
Name=Ravi
Gender=M
Height=5.800000
```